IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019-20

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

COMMUNICATION NETWORKS

Tuesday, 10 December 9:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : J.A. Barria
Second Marker(s) : C. Ling

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Special information for students

1. Mean delay for the M/M/1 system may be taken as

$$t_i = \frac{1}{\mu C_i - \lambda_i}$$

where,

$1/\mu$ = Average length of packet [bit/packet]

C_i = Transmission speed link i [bits/s]

μC_i = Service rate (link i) [packet/ s]

λ_i = Arrival rate (link i) [packet/ s]

2. Optimal Routing Problem (ORP)

$Min\ D(F)$ with respect to $F = \{F_i\}$

where, $D(F) = \sum_{i=1}^{L} \frac{F_i}{C_i - F_i}$

and,

C_i = Capacity of link l_i.

F_i = Flow carried by link l_i.

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The Questions

1.

a) In the context of Link layer ARQ protocols, assume that you know the transmission time of a single frame and the propagation delay of the medium.

i) Derive a simple expression for the following condition:

- The window of frames is transmitted in full before the acknowledgment of the first frame in the window is received by the sender. [2]

ii) For the condition stated in i) show that the link layer utilisation ratio of the *Selective repeat* ARQ protocol to the *Go back N ARQ* protocol can be approximated by the following expression:

$$\frac{U_{Selective\ repeat}}{U_{Go\ back\ N}} = 1 - P + WP \qquad [5]$$

Note: show step by step your derivations clearly also stating assumptions and approximations made.

iii) What is 1, P and W representing in this expression? [3]

b) Using the time line presented in *Fig.* 1.1. as a reference, for the CSMA/CD protocol:

i) Plot a typical sequence including successful transmission, collision/contention intervals and idle period. [2]

For the CSMA/CD protocol:

ii) Define a means to calculate the probability that exactly one station attempts transmission in one slot. [4]

iii) Derive an expression for the mean number of slots per contention interval. [4]



Figure 1.1:

2.

a)

i) Briefly state the background to the evolution of LTE (Long term evolution) and WiMax (Worldwide interoperability for microwave access) technology/standards. [4]

ii) Briefly state three similarities of the technological solutions of WiMax and LTE. [3]

iii) Briefly state three technology differences between WiMax and LTE. [3]

b)

i) State the optimality condition for the ORP of the network shown in Fig. 2.1. Assume that the cost associated to each link is modelled as the delay of an M/M/1 system. Where C and $3C$ are link capacities and the O-D demand is R. [4]

ii) Find the minimum value of R when all the links start carrying traffic. [3]

iii) Plot the optimal allocation of traffic curves: x_1^*, x_2^*, x_3^* vs R for the domain $R = (0, 3C)$, using *Fig.* 2.2 as a reference. [3]

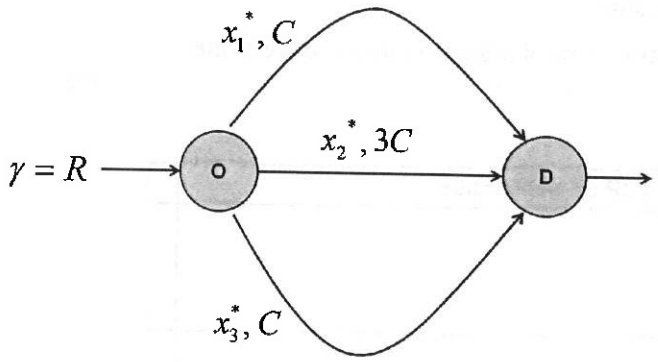


Figure 2.1:

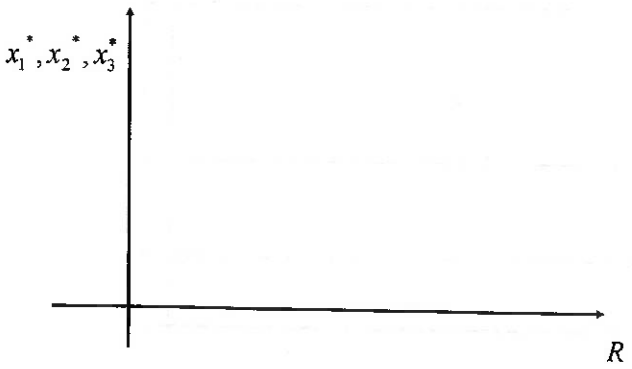


Figure 2.2:

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3.

a)

i) Briefly describe three elements that a host will require to connect to the Internet. [3]

ii) Describe briefly the Dynamic Host Configuration protocol (DHCP) operations. [3]

iii) Describe four elements of information that the DHCP server will inform a requesting DHCP client. [4]

b)

i) For the TCP congestion control algorithm, and using *Table* 3.1 as a reference, complete for each row the TCP sender actions. [5]

Note: assume initial value of the congestion windows *cwnd* is one *MSS* (maximum segment size), the threshold is large and that the sender is in slow start (*SS*) state.

ii) In the context of *TCP* latency estimation:

Derive the latency of the *TCP* congestion control assuming that you are at the beginning of a slow start (SS) state. [5]

State	Event	TCP Sender Action	
Slow Start (SS)	ACK receipt for previously unAcked data		
Congestion Avoidance (CA)	ACK receipt for previously unAcked data		
SS or CA	Loss event detected by triple duplicate ACK		
SS or CA	Timeout		
SS or CA	Duplicate ACK		

Table 3.1.

4.

a)

i) In the context of traffic shaping, briefly explain which traffic statistics could be monitored using a Leaky Bucket algorithm. [3]

ii) Using the sketch in *Fig.* 4.1. define the additional element(s) that need to be added to implement a Leaky Bucket algorithm. [3]

iii) Explain the mechanism of the Leaky Bucket in terms of its variables: bucket size B and token arrival rate D. [4]

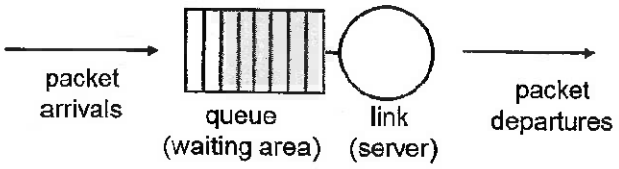


Figure 4.1.

b) The following play out time has been suggested for adaptive delay:

$$Play\ out\ time(i) = t_i + d_i + Kv_i$$

i) Define t_i, d_i, v_i and K. [4]

ii) Suggest how would you estimate d_i. [3]

iii) Suggest how would you estimate v_i. [3]